2025 International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN)
31 July - 2 August 2025, Rangpur, Bangladesh

GINet: A Deep Learning Approach to Gastrointestinal Image Classification

Provakar Mondol, Proma Das, and Kalyan Kumar Halder
Department of Electrical and Electronic Engineering
Khulna University of Engineering & Technology
Khulna-9203, Bangladesh
Email: provakarmondol24@gmail.com, promadaseee@gmail.com, kalyan@eee.kuet.ac.bd

***Abstract*—The gastrointestinal (GI) tract is one of the important parts of our body, performing the primary functions of digestion and absorption of food and nutrients. Many types of diseases hamper the normal operation of the GI tract and cause life-threatening conditions like cancer. Early detection of these diseases from endoscopic images is an essential step towards human well-being. A computer-aided system capable of automatically detecting diseases can reduce the pressure on doctors in underdeveloped countries who currently do this manually. This work proposes a newly modified deep convolutional neural network model, gastrointestinal network (GINet), which is trained to perform this task, achieving training and validation accuracies of 92.22% and 87.41%, respectively, on an 8-class dataset. The model was trained on the Kvasir V2 dataset, containing 8,000 images, with 1,000 images per class. Our goal is to classify GI images automatically with the least error. Based on the results, it is evident that this model is well-suited to handle GI tract data, enabling early detection of abnormalities.**

***Index Terms*—GINet, deep learning, endoscopy, colonoscopy, gastrointestinal tract, Kvasir V2 dataset.**

I. Introduction

The gastrointestinal (GI) tract begins at the oral cavity and extends through the esophagus, stomach, small and large intestines, concluding at the rectum and anus. It is a key part of our body where food and waste products are ingested, broken down, absorbed, and expelled. These vital organs can be affected by various diseases and other anomalies that impact the normal functioning of our digestive system. One of the most common types of disease is cancer, a life-threatening condition that can only be cured if detected early.

The World Health Organization (WHO) has declared colorectal cancer as one of the most life-threatening types of cancer. It accounts for 9.6% of the total death toll due to cancer worldwide, making it the 3rd highest among all cancers [1]. A survey by the WHO conducted in Bangladesh showed that cancer in the GI area had the highest contribution to the cancer population in 2022, accounting for about 15.1% of total cancer cases. It has taken the top position, surpassing breast cancer, which ranks 2nd on the list by a margin of almost 6% [2]. This clearly indicates the need to address abnormalities in the GI tract as early as possible to prevent them from developing into cancer, which would, in turn, reduce the mortality rate.

Endoscopy and colonoscopy are procedures used to detect abnormalities in the GI tract. These procedures use a camera and a tube that travel through the GI tract to capture images of existing conditions [3]. Some widely used tools for performing endoscopy include white light endoscopy and high-definition white light endoscopy, which are among the most commonly used.

It is evident that a great deal of laborious work is involved in the entire process [4]. In recent times, imaging quality has improved significantly, leading to better detection of anomalies in both endoscopy and colonoscopy. However, it remains a challenging and strenuous task for specialists and doctors to review each image and detect abnormalities, especially in a developing country like Bangladesh, where the pressure of increasing patient numbers grows daily. This often leads to erroneous detection, even after taking necessary precautions, which cannot always be avoided. This situation highlights the need for computer-aided detection of abnormalities in endoscopic and colonoscopic images. Such technology alleviates the workload of specialists and reduces the time required to analyze data from a single patient, enabling hospitals and diagnostic centers to better handle the growing patient demand.

Since the evolution of neural networks, researchers have focused on finding more efficient ways to detect images using convolutional neural networks (CNNs). However, there are still constraints with deep learning models, as they require large datasets for effective training. Additionally, with high-quality images and videos, the data size increases significantly, making it challenging to train any network. Thousands of annotated images are needed to make a model fully reliable and capable of replacing the manual detection system.

In this work, we propose an improved classification model, named the gastrointestinal network (GINet), based on CNN, which can distinguish among various images obtained from a patient's endoscopy and colonoscopy. We used the Kvasir V2 dataset [5] to train our model and tested it with the validation set of data. Due to the data scarcity issue, augmentation was also applied to the model, and results were compared for augmented and non-augmented data. We also applied several pre-trained models to the same dataset and evaluated their metrics to compare them with our model. The accuracy, precision, recall, and F1-score were the metrics used for this comparison, showing that our model performs well in detecting the classes in colonoscopy and endoscopy images while having fewer parameters than other models.

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II. Literature Review

The past few years have seen significant advancements in neural networks, particularly in CNNs. Alongside these developments, research in detecting abnormalities in the GI tract has also progressed considerably, and this trend continues. Over time, detection systems are becoming increasingly efficient.

Many research groups have explored various techniques. Demirbas et al. [3] proposed a novel architecture based on a spatial-attention ConvMixer and compared its performance with state-of-the-art models commonly used to detect GI abnormalities. Ghatwary et al. [4] proposed a faster region-based CNN model to detect abnormalities in the esophagus, achieving an F1-score of 75%. Their research was limited to normal esophagus and esophagitis data from the Kvasir dataset. Similarly, Trinh and Nguyen [6] presented a model combining two pre-trained networks, DenseNet121 and ResNet101, to detect GI tract anomalies, utilizing high-end GPUs to meet significant computational requirements. Hosain et al. [7] employed a transfer learning technique based on the DenseNet201 and Vision Transformer architectures to differentiate between normal colon images and three GI disorders: ulcerative colitis, polyps, and esophagitis. The top layer of DenseNet201 was removed as the study was limited to a four-class classification task.

Khan et al. [8] presented a fully automated method based on a pre-trained VGG16 model, where annotations were manually provided for ulcer images. A commonly used detection system for adverse events in GI endoscopy, called the AGREE system, was proposed by Nass et al. [9]. However, this classification lacks subjective input from the global survey presented in their report. Obayya et al. [10] proposed a modified salp swarm algorithm for endoscopic image classification of GI tract diseases, in which they adapted the CapsNet model for feature extraction to classify wireless capsule data from the GI tract.

Sharma et al. [11] developed a custom U-Net model to detect organs in GI tract images and cross-validated its performance against several pre-trained models, including Inception V3, VGG19, ResNet50, and DenseNet, to demonstrate its superior effectiveness over these existing models. Elmagzoub et al. [12] enhanced the performance of various pre-trained CNN models commonly used for medical image classification by adjusting hyperparameters in their study. Gunasekaran et al. [13] employed a weighted average ensemble CNN model to classify the Kvasir dataset and compared its performance to that of individual models; however, the validation accuracy and loss curve lacked stability. Both proposed models are highly complex and require high-end hardware setups.

III. Methodology

In this section, we present a modified CNN model named GINet to classify the images of the GI system. The diagram in Fig. 1 illustrates the steps of this entire process. The first stage involves dataset creation, reshaping, one-hot encoding, and augmentation, which are the preprocessing steps required for the images to be fed into our model. Next, we built and trained the model using the training data. We modified the convolution layers, batch normalization layers, dropout layers, and dense layers to optimize the model's output. We recorded the results for each training and testing phase, and then prepared the model for further training by adjusting the layer combinations and hyperparameters in such a way that they are designed to achieve the best metrics and optimization on a trial-and-error basis. Later, we applied the model to non-augmented and augmented data for comparison. Using these hyperparameters, we tested other pre-trained models for comparison and recorded the results.

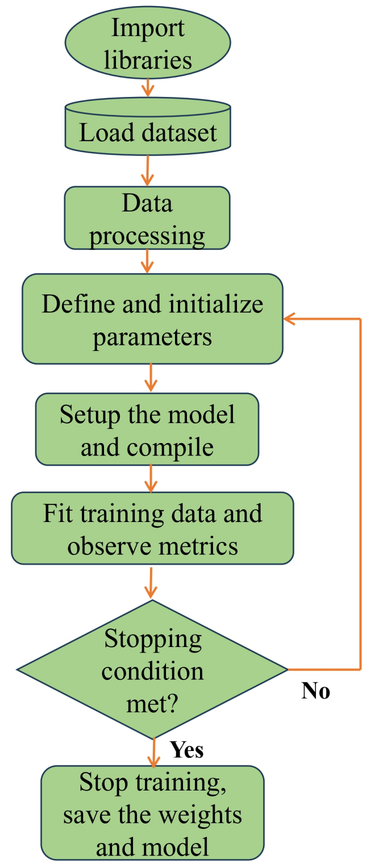


Fig. 1. Flow chart of the CNN model.

A. Proposed Architecture of CNN Model: GINet

Our deep learning-based CNN consists of a combination of 2D convolutional blocks along with a normalization layer and a pooling layer. Finally, the fully connected layer is used, which enables automatic feature extraction, allowing the model to detect images on its own. The conventional CNN model was modified to have the least number of parameters with the highest accuracy. The total number of parameters was 59,03,816. The configuration of our proposed GINet is shown in Table I, and the basic building blocks used for the model are also described accordingly.

TABLE I
CONFIGURATION OF PROPOSED MODEL

Layers	Output Shape	Parameters
Conv2D_1 (activation:ReLU)	(None, 198, 198, 32)	896
BatchNormalization_1	(None, 198, 198, 32)	128
MaxPooling_1	(None, 99, 99, 32)	0
Conv2D_2 (activation:ReLU)	(None, 97, 97, 64)	18496
BatchNormalization_2	(None, 97, 97, 64)	256
MaxPooling_2	(None, 48, 48, 64)	0
Conv2D_3 (activation:ReLU)	(None, 46, 46, 128)	73856
BatchNormalization_3	(None, 46, 46, 128)	512
MaxPooling_3	(None, 48, 48, 64)	0
Conv2D_4 (activation:ReLU)	(None, 21, 21, 256)	295168
BatchNormalization_4	(None, 21, 21, 256)	1024
MaxPooling_4	(None, 10, 10, 256)	0
Conv2D_5 (activation:ReLU)	(None, 8, 8, 512)	1180160
BatchNormalization_5	(None, 8, 8, 512)	2048
MaxPooling_5	(None, 4, 4, 512)	0
Flatten	(None, 8192)	0
Dense_1 (activation:ReLU)	(None, 512)	4194816
BatchNormalization_6	(None, 512)	2048
Dropout_1 (0.2)	(None, 512)	0
Dense_2 (activation:ReLU)	(None, 256)	131328
BatchNormalization_7	(None, 256)	1024
Dropout_2 (0.2)	(None, 256)	0
Dense_3 (activation: softmax)	(None, 8)	2056

1) Input layer: We reshaped all the images in our dataset to 200×200 RGB images and fed them into the input layer.

2) Convolution layer: Five convolution layers constitute the basic building block of our proposed model, each with a kernel size of 3×3 and filter sizes ranging from 32 to 512.

3) Activation layer: Our model uses ReLU as the activation function after each convolutional layer.

4) Normalization layer: Batch normalization is applied after every convolutional layer and its corresponding activation layer. It helps maintain uniformity in our dataset.

5) Pooling layer: Max pooling with a kernel size of 2×2 is used in our model to reduce the number of features after each convolution.

6) Dense layer: Finally, three dense layers are used, with the final output layer having 8 neurons, as our output consists of 8 classes by design. A dropout layer was also used to avoid overfitting. The activation function for the final layer was softmax.

B. Other Transfer Learning Models

We have considered some pre-trained models to compare with the proposed GINet on the same dataset with the same hyperparameters and image size. The models are discussed briefly below.

1) DenseNet: Huang et al. [14] developed DenseNet that shares the same feature map size directly between any layers, and achieved a significant improvement in accuracy and error rates on datasets like CIFAR, SVHN, and ImageNet.

2) Inception V3: Szegedy et al. [15] established a finer model in continuation of Inception V2, which works very well on low-resolution single-frame image dataset classification. It also recorded a lower error rate than contemporary models.

3) EfficientNet: Tan and Le [16] proposed a new scaling method that balances the width, depth, and resolution of image dataset and achieved an accuracy of 84.1% on ImageNet by engaging an effective compound coefficient.

4) ResNet: He et al. [17] proposed an efficient deep learning-based CNN model called ResNet, known for its use of shortcut connections. In the term ResNetX, "X" refers to a residual deep neural network with X layers. ResNet50 and ResNet101 are among the most popular models in this series.

5) MobileNet: Sandler et al. [18] proposed a CNN model called MobileNet. The model uses depth-wise separable convolution layers, which are a combination of two 1D convolution layers with different kernels. This model requires less memory and fewer parameters, providing efficient results from a relatively small model.

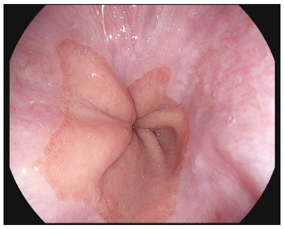
IV. RESULTS AND DISCUSSION

This section describes the results obtained by our model and other already developed models in terms of various neural network metrics. It also describes the hardware machine setup and details on the dataset.

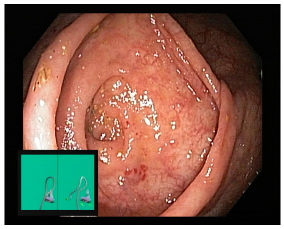
A. Dataset Details

We used the Kvasir V2 dataset [5] for our training, which consists of 1,000 images in each of the 8 classes, totaling 8,000 images. The dataset was collected using endoscopes and colonoscopes at Vestre Viken Health Trust in Norway. Among the 8 classes of images, 3 represent anatomical landmarks, 3 represent pathological findings, and 2 represent stages of polyp removal. Sample images from this dataset are shown in Fig. 2.

We split the dataset into training and validation sets, allocating 80% and 20% of the data, respectively. The validation set was used for testing the performance of the model. This dataset is considered small for deep learning models. To overcome this shortcoming, we further augmented the dataset, preparing it for real-world image analysis. We used the ImageDataGenerator function from Keras for data augmentation, applying transformations such as rotation, horizontal flipping, zoom, width shift, and height shift. These augmentations were applied only to the training data to help the model generalize better. The validation set was not augmented to maintain consistent evaluation criteria, as this was used for testing our model. This approach enabled the model to train on a more diverse dataset, ultimately improving its accuracy, which was proven by the accuracy we achieved during training and testing.

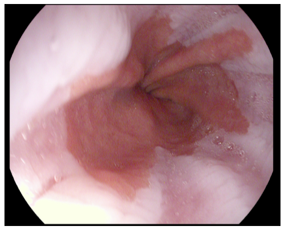


(a) Normal Z-Line



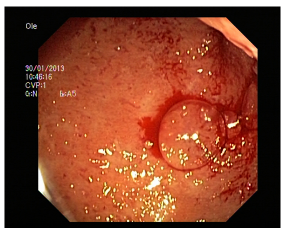
(b) Normal Cecum

(c) Pylorus

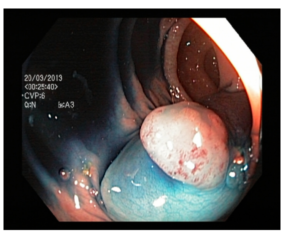


(d) Esophagitis

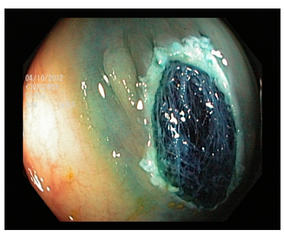
(e) Polyps



(f) Ulcerative Colitis



(g) Dyed Lifted Polyp



(h) Dyed Resection Margin

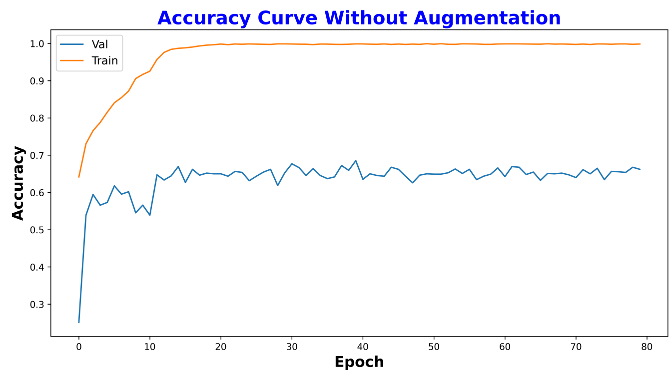
Fig. 2. Sample images from Kvasir V2 dataset [5].

B. Hardware Setup

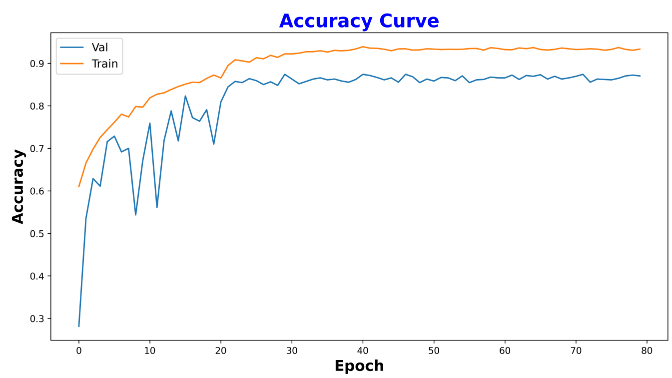
The Python programming language was the primary medium for implementing our proposed model using Anaconda Navigator. In Anaconda Navigator, we created an environment with a TensorFlow (v2.10.0) background and Keras as part of the TensorFlow library, which we accessed through Jupyter Notebook. Our hardware setup consisted of a central processing unit with an Intel Core i5 8th Gen (8265U). The processor speed was 1.60 GHz (with a maximum of 1.80 GHz). The operating system used was Windows 10 Pro with an OS build of 19045.5011, 8 GB RAM, a 256 GB solid-state drive, and a 1 TB hard drive. This setup promises a viable solution to real-life applications as the model is lightweight and requires less computational capability.

C. Accuracy and Loss Curves

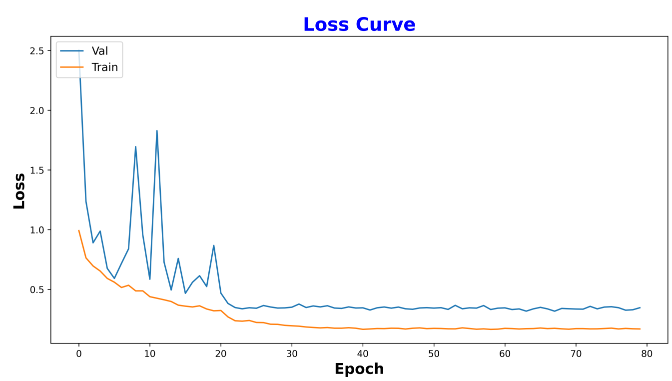
Initially, the model was trained using non-augmented data to evaluate its baseline performance. As shown in Fig. 3(a), the resulting accuracy curve exhibits abrupt fluctuations, with a relatively low validation cum testing accuracy of 67.32%. This behavior indicates a tendency toward overfitting. To address this issue, data augmentation was applied. The corresponding accuracy and loss curves after augmentation are presented in Fig. 3(b) and Fig. 3(c), respectively. These plots demonstrate that both accuracy and loss gradually stabilized after a certain number of epochs, effectively mitigating the limitations observed with the non-augmented dataset. There is a nominal gap between the training and validation accuracies as well as between the training and validation losses. This is because the validation data were never exposed to the model during training and were only used for validation of the model.



(a) Accuracy curve without augmentation



(b) Accuracy curve with augmentation



(c) Loss curve with augmentation

Fig. 3. Accuracy curves and loss curves during training and validation for the proposed GINet model.

One noticeable point in this work is the optimizer and callback functions used for the models. Adam was used to optimize all models, and we used a learning rate reducer as a callback function to save computational power, cost, and reduce the training time to just 297 minutes. The learning rate reducer relied on validation accuracy for decision-making. From the curves, it is clear that the model reached its global minimum at approximately 50 epochs, with stabilization observed after 30 epochs. Also, the time required for validation cum testing was significantly lower for this modified model, taking only 9 seconds, whereas other models took considerably longer to make predictions.

D. Confusion Matrix

The confusion matrix is a representation of the overall model's accuracy. It is plotted based on the predictions made by the model after training is completed, using the test dataset, which the model has not seen until this stage. We have plotted the normalized confusion matrix for our model and other models. The normalized confusion matrix for our model, shown in Fig. 4, demonstrates that it can accurately predict the class of any given test image after training. However, some misclassifications occurred between the "normal z-line" and "esophagitis" categories, likely due to similar color profiles and anatomical areas of the GI tract shared by these two image classes, as can be seen in Fig. 2. Our model performed quite well, predicting classes of eight random images correctly in each round, which can also be seen in Fig. 5.

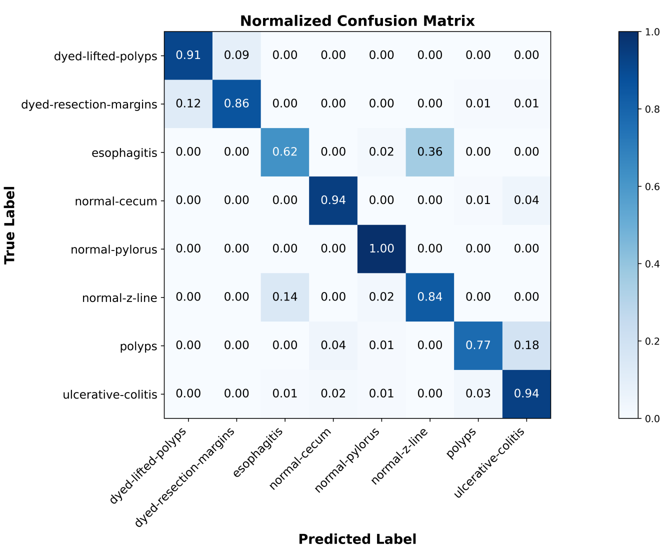


Fig. 4. Normalized confusion matrix.

E. Evaluation Metrics

All models were implemented on the same dataset and with the same data augmentation setup. The number of epochs, batch size, loss function, optimizer, and learning rate were also kept the same so that all models followed the same base learning configuration. The metrics used for this comparison are listed below.

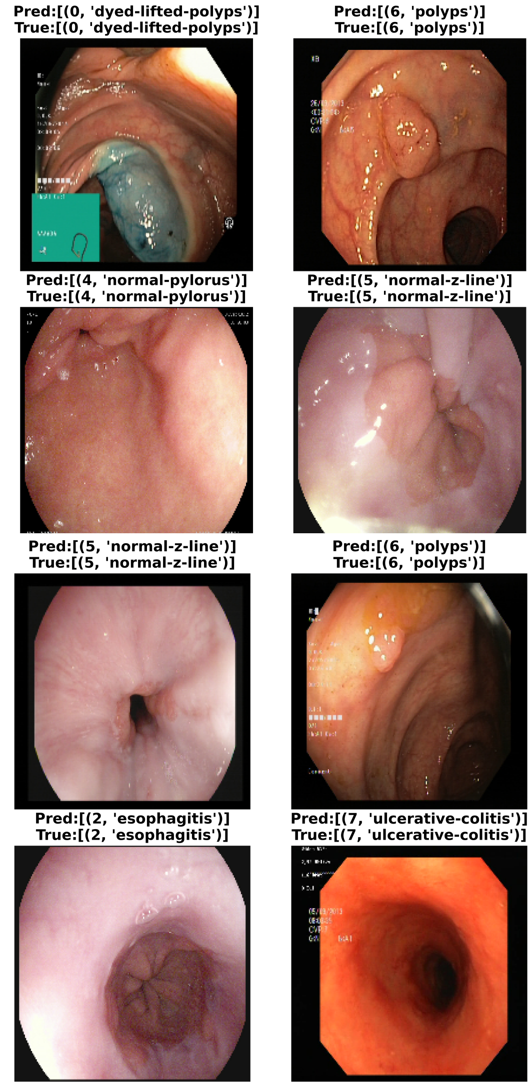


Fig. 5. Sample prediction results on test images.

1) Accuracy: This is a measure of how accurately a model can classify images after training in terms of True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN), as defined in (1). It indicates how accurately our model can perform on a new dataset, with an accuracy of 92.22% for the training dataset and 87.41% for the validation dataset when trained with augmentation.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \tag{1}$$

2) Precision: Equation (2) represents the precision metric, which measures how many TPs a model correctly identifies out of all predicted positive samples. Our proposed model achieved a precision of 86.65%.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{2}$$

3) Recall: Recall is a measure of how many positive samples are correctly identified by the model out of all the

TPs and FNs in the dataset. For the model we developed, the recall was 86.36%, as calculated using the formula in (3).

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{3}$$

4) F1-score: The F1-score is calculated as the harmonic mean of recall and precision, as shown in (4). The F1-score for our proposed model was 86.15%, which outperformed the other models we tested for transfer learning.

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

After evaluating each model, including our proposed model with augmentation, we recorded the metric scores to facilitate comparison. The experimental results are shown in Table II. This comparison reveals that our proposed model achieves higher accuracy in the validation cum testing phase. Additionally, other metrics are also higher for our model. The number of parameters and time required for decision-making were both lower in our case, which makes our modified model stand out.

TABLE II
METRICS COMPARISON AMONG DIFFERENT MODELS

Models (Parameter Count)	**Evaluation Metrics**			
	Accuracy	**Precision**	**Recall**	**F1-score**
DenseNet (12.8 Million)	84.32%	78.97%	79.13%	79.44%
Inception V3 (23.8 Million)	81.52%	80.11%	79.92%	80.21%
EfficientNet (26.1 Million)	78.56%	77.46%	78.12%	78.71%
ResNet50 (23.9 Million)	83.57%	81.45%	81.02%	80.99%
MobileNetV2 (7.4 Million)	72.09%	71.67%	71.28%	71.21%
Proposed Model (GINet) (5.9 Million)	87.41%	86.65%	86.36%	86.15%

V. CONCLUSION

This paper presents the results of a modified deep learning-based CNN, GINet, and compares its performance with several pre-trained models with the same computational setup and set of hyperparameters. Our model demonstrates superior performance in classifying the augmented Kvasir V2 dataset, achieving a validation accuracy of 87.41%. The precision, recall, and F1-score values are also very satisfactory, at 86.65%, 86.36%, and 86.15%, respectively. Augmentation helped us with data scarcity greatly. Having fewer parameters than the state-of-the-art models marked the simplicity of our model. Additionally, our model is optimized to run efficiently on a moderate computational setup with lower time requirements, reducing overall computational costs.

Despite its promising results, the proposed GINet model faces limitations such as the use of a relatively small and homogeneous dataset, which may restrict its generalizability in diverse clinical settings. Additionally, the absence of real-time testing, clinical validation, and integration into existing healthcare workflows poses significant challenges to its adoption in practical medical environments. Regulatory compliance and explainability requirements further add to the barriers for deployment in clinical practice. Looking ahead, we plan to develop a hybrid deep learning model to further enhance performance metrics for image classification, along with a graphical user interface. We also aim to design a new deep learning model for real-time video data analysis using the Hyper Kvasir dataset.

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